Fiscal Year:	FY 2013	**Task Last Updated:**	FY 02/12/2013
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Levitation Observation of Dendrite Evolution in Steel Ternary Alloy Rapid Solidification (LODESTARS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Tufts University		
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PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	04/03/2008	**End Date:**	04/02/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	8	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 4/2/2014, per P. Downey/MSFC (Ed., 2/21/2012) NOTE: Received extension until 4/02/2012 per NSSC (Ed., 4/15/2011) NOTE: Grant may be extended, per P. Downey/MSFC (Ed., 2/18/2011)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Hyers, Robert (University of Massachusetts, Amherst)		
Grant/Contract No.:	NNX08AL21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Undercooling in containerless processing has shown promise in producing metastable phases and materials that exhibit improved chemical homogeneity and ultra-fine grain sizes. Stainless steel alloys require special handling in order to achieve a significant undercooling range. In particular, chromium is particularly susceptible to contamination through reaction with crucible material. When attempting to perform rapid solidification tests, the formation of active heterogeneous nucleation sites limits the undercooling that may be achieved. By using a containerless technique, this contamination is minimized. The focus of this program is to study of the effect of microgravity on metastable solidification. By observing changes in the mechanism for nucleation of the stable phase following primary recalescence from undercooled melts, the role of convection in phase selection may be evaluated. The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Conducted flight support ESL density tests at ESL-MSFC 6-17 August 2012 Conducted flight support ESL viscosity tests at ELS-MSFC 7-18 January 2013 Conducted ground-based test sample evaluation of samples produced at the MSFC ESL in collaboration with MSFC M&P metallography staff Release of Solidification of Containerless Undercooled Melts book edited by DM Herlach and DM Matson Graduated three MS students International Working Group 15-16 November 2012 in Friedrichshafen DE. IWG 7 involved evaporation limit planning and payload status reporting. Visit to view flight hardware development by Astrium Conducted Parameter Set definition 28 Jan – 1 Feb 2013 by MUSC at DLR Köln DE for FeCo flight alloy samples 6 and 7.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Matson DM. "Materials science in reduced-gravity." JOM (Journal of The Minerals, Metals & Materials Society (TMS) 2012 Sep;64(9):1087-8. http://dx.doi.org/10.1007/s11837-012-0425-y , Sep-2012
Articles in Peer-reviewed Journals	Matson DM. "Items of Note from the Field, Profession, and Society: Materials Science in Microgravity." JOM (Journal of The Minerals, Metals & Materials Society (TMS) 2012 Jun;64(6):625. , Jun-2012
Books/Book Chapters	Herlach DM, Matson DM. Editors. "Solidification of Containerless Undercooled Melts." Ed. D.M. Herlach, D.M. Matson. Hoboken, N.J. : Wiley-VCH, 2012., Aug-2012
Books/Book Chapters	Matson DM. "Chapter 10 - Nucleation within the Mushy Zone." in "Solidification of Containerless Undercooled Melts." Ed. D.M. Herlach, D.M. Matson. Hoboken, N.J. : Wiley-VCH, 2012., Aug-2012
Books/Book Chapters	Hyers RW. "Chapter 2 - Computer aided experiments in containerless processing of materials." in "Solidification of Containerless Undercooled Melts" Ed. D.M. Herlach, D.M. Matson. Hoboken, N.J. : Wiley-VCH, 2012., Aug-2012
Dissertations and Theses	Choufani P. "Measuring the thermophysical properties of Fe-Co using containerless electrostatic levitation techniques." MS-thesis, Tufts University, Medford, MA, August 2012. , Aug-2012
Dissertations and Theses	Dolan J. "Analysis of the adiabatic remelt model as applied to the solidification of undercooled iron-cobalt alloys." MS-thesis, Tufts University, Medford, MA, May 2012. , May-2012
Dissertations and Theses	Sander P. "Characterizing the performance of an induction-heated molten oxide reactor used for electrowinning of iron." MS-thesis, Tufts University, Medford, MA, May 2012. , May-2012
NASA Technical Documents	Hyers R, Bojarevis V, Downey J, Henein H, Matson D, Seidel A, Voss D, Eds., and SanSoucie M, compiler. "Materials Research in Microgravity 2012. Proceedings of a symposium held at the 141st TMS 2012 Annual Meeting and Exhibition, Orlando, Florida, March 11–15, 2012." Huntsville, Ala. : George C. Marshall Space Flight Center, 2012. NASA Conference Publication NASA/CP-2012-217466. http://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20120015582_2012018023.pdf , Sep-2012
NASA Technical Documents	Lee J, Choufani P, Bradshaw RC, Hyers RW, Matson DM. "Containerless Measurement of Density and Viscosity in Fe-Co Alloys." In: Materials Research in Microgravity 2012. NASA conference publication NASA/CP-2012-217466, 2012. p. 97-104. http://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20120015593_2012018063.pdf , Sep-2012
NASA Technical Documents	Matson DM, Sander PJ. "Truncated dual-cap nucleation site development." In: Materials Research in Microgravity 2012. NASA conference publication NASA/CP-2012-217466, 2012. p. 193-200. http://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20120015604_2012018049.pdf , Sep-2012